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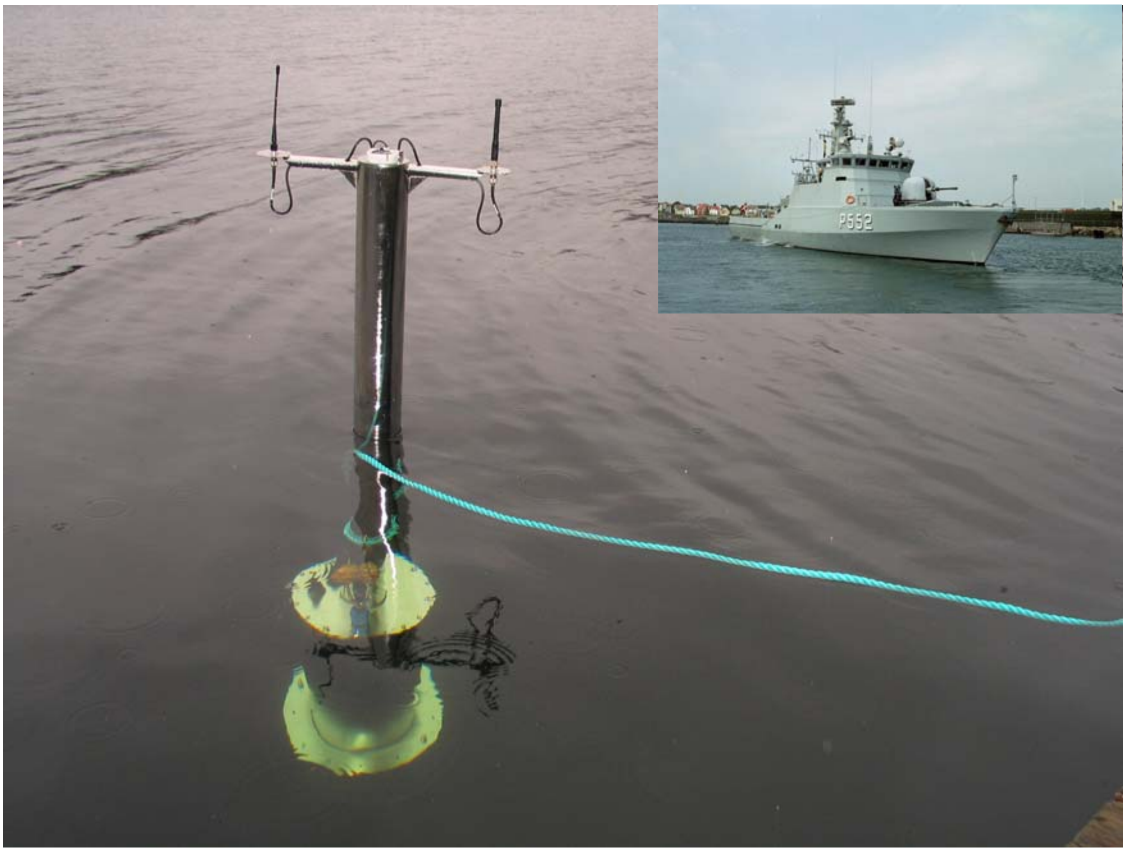

TMK-SAS

(Sound Registration and Analyzing system)

Applications

- *Mapping of noise signature for ships and marine installations*
    - *Mine sweepers - to ensure safety distance*
    - *Fishing vessels - during search and catch operation*
    - *Seabed installations – to detect malfunction*
- *Biologic noise in the ocean including mammal sound*
- *Determination of Shock wave sources.*

The ***Sonar Registration and Analyzing System (TMK-SAS)*** is a complete system for receiving, storing and analyzing underwater sound.

The system consists of two main parts:

- A GPS positioned, battery operated buoy, containing all necessary electronics for amplifying signals from the hydrophone, A/D converting, storing, analyzing in frequency bands and transmitting data to the Control Station (SCS).
- A PC-based, GPS positioned Control Station (SCS) on board the ship or offshore installation for control of buoys settings and presentation of received data.

Figure 2 TMK-SAS

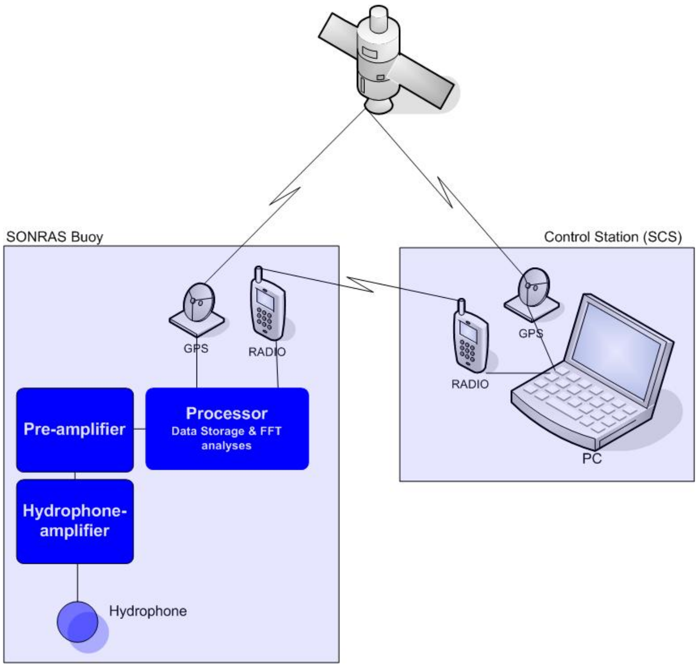

Figure 1 System Overview

The buoy records sound from the sea, analyses and transfers the result to the SCS. Raw data and processed data are stored on CompactFlash in the buoy for later analysis.
A complete plot of GPS positions is also stored in the buoy, so that relative distance can be computed in the SCS for normalizing the noise data.

The Buoy receives setup commands from the Control Station (SCS), and reports status messages and measurements back over the UHF data link modem.
Low noise hydrophone preamplifiers together with 16 bits A/D converters and controlable amplifiers gives a total dynamic range from ambient noise level up to very noisy test objects.

In operation the TMK-SAS buoy performs the following tasks:

- Read and report the GPS positions.
- Signals received from the hydrophone are continuously sampled (42 Ks/sec) for storing and analyzes in the frequency band from 3 Hz to 20 kHz.
- The frequency band from 20 kHz to 100 kHz is analyzed in numerical filters.
- The SCS is updated every 1.5 sec with 1/3 octave band data in the full frequency range 3Hz to 100kHz
- (Option) In mode for shock wave or splash down source positioning, events are time tagged to GPS time with accuracy better than 1 msec.
- Audio signal (“listen in”) is obtained by sending the analogue signal over a VHF radio link (frequency range approx. 100 Hz to 3 kHz).
- Both the data stored in the buoy and the data transmitted to SCS by radio is encrypted.

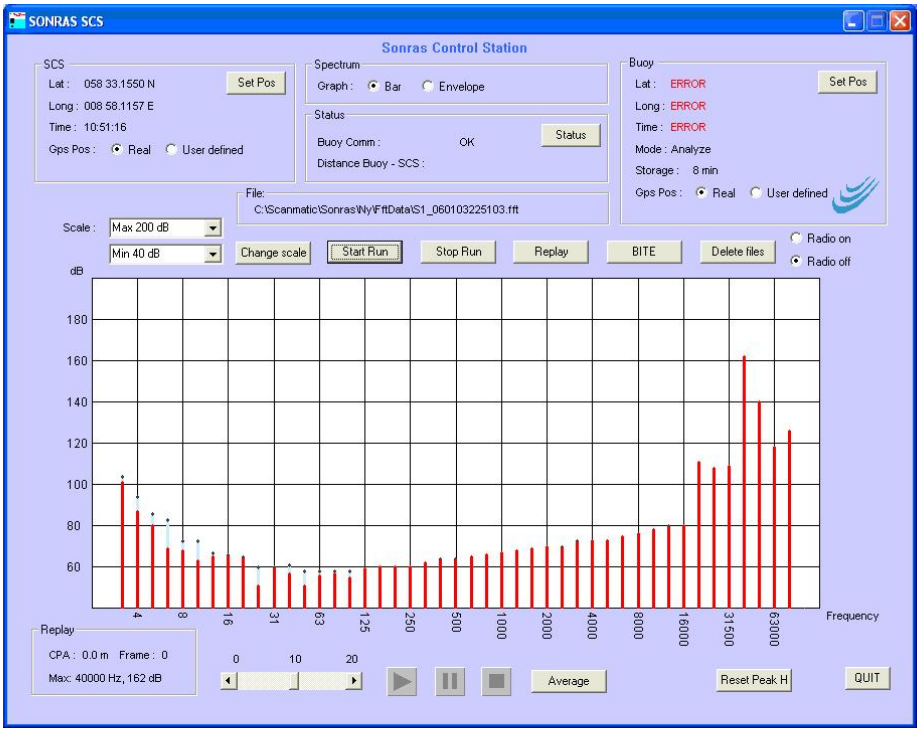

Figure 3 Control Station (SCS) User Interface

The ***Control Station (SCS)*** has 4 modes of operation:

- Control and monitoring the of data collection. During this operation, noise data analyzed to 1/3 octave band is displayed every 1.5 sec, reduced to true spectrum levels. The analogue sound can be listened to by activating the VHF radio.
- Replay of received processed data together with the recorded analogue signals.
- (Option) Positioning of events. The SCS can receive time-tagged events from several buoys for positioning of sound source.
- Post-processing of raw data from the buoy. The signals sampled and stored in the buoy can be retrieved from the buoy by Ethernet connection after the measurement mission is finished and the buoy is recovered. This enables new FFT analyzes in the band from 3 Hz to 20 kHz now also with narrow band analyzes (resolution 1.5 Hz)

Figure 4 Control Station (SCS)